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Roll No

FT-405 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Pumping Machinery and Fluid Mechanics
Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Define slip and percentage slip of a reciprocating pump. How will you classify the reciprocating pumps?
b) A double-acting reciprocating pump, running at 40 r.p.m., is discharge 1.0 m^3 of water per minute. The pump has a stroke of 400 mm. The diameter of the piston is 200 mm. The delivery and suction head are 20 m and 5 m respectively. Find the slip of the pump and power required to drive the pump.
2. a) State and prove Pascal's Law.
b) An inverted differential manometer containing an oil of sp.gr. 0.9 is connected to find the difference of pressures at two point of a pipe containing water. If the manometer reading is 40cm. Find the difference of pressures.
3. Water is flowing through a pipe of diameter 30cm at a velocity of 4m/s. Find the velocity of oil flowing in another pipe of diameter 10cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water and oil is given as 0.01 poise and 0.025 poise. The specific gravity of oil = 0.8.
4. a) What are the methods of dimensional analysis? Explain the Buckingham- π theorem.
b) Water is flowing through a pipe of diameter 30 cm at a velocity of 4m/s. Find the velocity of oil flowing in another pipe of diameter 10 cm, if the condition of dynamic similarity is satisfied between the two pipes. The viscosity of water and oil is given as 0.01 poise and 0.025 poise. The specific gravity of oil = 0.8.
5. An inverted differential manometer containing an oil of sp.gr. 0.9 is connected to find the difference of pressures at two point of a pipe containing water. If the manometer reading is 40cm. Find the difference of pressures.
6. The outer diameter of an impeller of a centrifugal pump is 400 mm and outlet width is 50 mm. The pump is running at 800 r.p.m. and is working against a total head of 15 m. The vanes angle at outlet is 40° and manometric efficiency 75%. Determine :
 - i) velocity of flow of outlet,
 - ii) velocity of water leaving the vane,
 - iii) angle made by the absolute velocity at outlet with the direction of motion at outlet, and
 - iv) discharge.

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7. What do you mean by priming of a centrifugal pump? Explain how it is done?

OR

A centrifugal pump, having impeller diameter of 250mm and 500mm at inlet and outlet respectively, is running at 800 rpm. The vanes are set back at an angle of 30° to the outer rim. If the velocity of flow through the impeller is constant at 2m/s, find :

- i) Vane angle at inlet,
 - ii) Work done per kg mass of water on the wheel.
8. a) Define the terms:
- i) Path line and Stream line
 - ii) Reynold's number and Froude number.
- b) A pipe carrying oil of specific gravity 0.87, changes in diameter from 200 mm diameter at apposition A to 500 mm diameter at a position B which is 4 metres at a higher level. If the pressures at A and B are 9.81 N/cm^2 and 5.886 N/cm^2 respectively and the discharge is 200 liters/s. Determine the loss of head and direction of flow.

OR

- a) What do you mean by rate of flow or discharge? Distinguish between rotational and irrotational flow.
- b) If for a two-dimensional potential flow the velocity potential is given by $\Phi = x(2y - 1)$. Determine the velocity at the point P (4, 5). Determine also the value of stream function Ψ at the point P.
